

MOHAMMAD KAMAL ABDULAZIZ

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SUMMARY

Data Scientist and AI Engineer (B.Sc. Data Science, University of Jordan) working on computer vision, multimodal AI, MLOps and applied machine learning. Experience with VLM spatial reasoning, medical image classification, object detection and segmentation, and calibration. Published research in medical image classification; uses PyTorch, FastAPI, Docker and cloud.

EDUCATION

The University of Jordan

Amman, Jordan

- **B.Sc. in Data Science**, King Abdullah II School of Information Technology | **Graduated Jan 2026**.

EXPERIENCE

Sager Drone

Jul 2026 – Present

AI Engineer Intern

- **VLM Spatial Reasoning & RAG (Banan Site)**: Developed a deterministic Q&A engine using **Qwen2.5-VL**. Engineered a RAG pipeline powered by a Knowledge Graph and dual-database architecture (spatial index and domain knowledge). This system dynamically queried and expanded the VLM's prompt context and verified responses through an algorithmic gate.
- **Perception AI Pipeline**: Built a React/FastAPI MLOps platform for training vision models, featuring automated calibration metrics, hardware-aware batching, and group-aware cross-validation to prevent data leakage.

University of Jordan & Akdeniz University

Feb 2026 – Jun 2026

Research Assistant (AI Modeler) | Joint Research Project (Remote)

- Curated binary and multi-class clinical image datasets from the ISIC archive to support multi-tier diagnostic workflows.
- Trained two complementary architectures: a lightweight edge model for binary classification and a higher-capacity model for multi-class classification.

Globitel

Nov 2025 – Dec 2025

AI Engineer

- Managed remote model-training workflows using **Termius** and validated model updates for enterprise AI services.
- Built automated data-preparation pipelines for labeling, organization, and train/validation/test splitting.
- Upgraded classification and emotion detection models, validating performance via **Postman** API testing.

Al Ain University Student Research Conference

Jan 2026

International Presenter (UAE): presented AI and data science research

PUBLICATIONS

GISLC: Gated-Inception Model for Skin Lesion Classification

Feb 2026

mdpi.com/2079-9292/15/4/861

- Peer-reviewed in **Electronics (MDPI)**, 2026. Co-designed the Gated-Inception architecture and its spatial gating mechanism for adaptive feature fusion; 98.23% accuracy and 98.23% F1 in the reported evaluation.

PROJECTS

APEX: Latent-Space Kernel Fusion for Selective Prediction

Jun 2026

github.com/Mohammad-Kamal23/apex-selective-prediction

- Post-hoc kernel fusion layer for frozen classifiers, evaluated on 18 configurations (6 medical imaging modalities $\times$ 3 backbones).
- AURC -44.3% and balanced accuracy $+4.18$ points; beats 6 baselines in 18/18 configurations ($p = 7.6 \times 10^{-6}$).

Automated Document Stamping (n8n + Vision AI)

Apr 2026

github.com/Mohammad-Kamal23/n8n-sign-project

- **Automated pipeline**: containerised **n8n** and **FastAPI** services that apply digital approvals to bilingual documents.
- **Fusion engine**: deterministic OpenCV matching merged with zero-shot **Florence-2** grounding, without double-stamping.

CropEye: Smart Agriculture Assistant (in development)

May 2025 – Jan 2026

github.com/Mohammad-Kamal23/CropEye

- Full-stack system: **Flutter** image streaming, **FastAPI** inference backend and a **Gemini**-powered agronomy assistant.
- **YOLOv8** leaf detection and lightweight **ShuffleNetV2** damage grading (healthy, mild, moderate, severe).

TECHNICAL SKILLS & EXPERTISE

Core: Python, PyTorch, FastAPI, Docker, Git, Computer Vision, Deep Learning.

AI: VLMs, RAG, Object Detection, Segmentation, Image Classification, Calibration & Selective Prediction.

Cloud & Deployment: GCP, AWS, REST APIs, ML pipelines, MLOps.

Data & Geospatial: SQL, NoSQL, GeoJSON, QGIS, ETL/ELT.

Languages: Arabic (Native), English (Fluent).